Cantera vs NN | Test IC 05 | NRMSE [%]

T0 = 1049.0 K P0 = 4.666 atm phi0 = 0.674 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE}(\%) = 100\,\mathrm{RMSE}_{time}/(\max_t y_{Cantera} - \min_t y_{Cantera})$$

Model / same-state mean	Training	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	43.9	43.5	2.94	11.3	54.9	501	42.3	11.2	14.7	N/A	Complete
2 P-only DeepONet10	**10k**	83	83	2.76	11	83	44	83	10.8	14.6	N/A	Complete
2 P-only DeepONet10	**long**	88.2	80.3	5.2	92	73.9	22.7	81.8	61.5	14.2	N/A	Complete
3 ANN10 + P-only DeepONet10	**10k**	83	83	2.76	11	83	44	83	10.8	14.6	N/A	Complete
3 ANN10 + P-only DeepONet10	**long**	39.8	42.2	51.6	9.03	41.6	53	40.7	10.7	14.6	N/A	Complete
4 T/P DeepONet10	**10k**	51.5	53	8.07	9.78	54	58	52	10.7	14.6	N/A	Complete
4 T/P DeepONet10	**long**	80.5	54.2	147	8.09	83.9	1.1e+03	55.6	10.9	14.6	N/A	Complete
5 ANN10 + T/P DeepONet10	**10k**	50	51.4	5.78	8.03	52.4	58.8	50.4	9.98	14.5	N/A	Complete
5 ANN10 + T/P DeepONet10	**long**	55.1	50.2	46.5	10.7	73.2	825	47	9.69	14.5	N/A	Complete
6 T/P/ϕ DeepONet10	**10k**	83	83	2.76	11	83	44	83	10.8	14.6	N/A	Complete
6 T/P/ϕ DeepONet10	**long**	62.7	49.4	188	6.52	58	575	45.8	83.3	14.1	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	83	83	2.76	11	83	44	83	10.8	14.6	N/A	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	44.8	49.8	84.1	8.55	56	370	43.3	22.6	14.5	N/A	Complete

RMSE uses trapezoidal time integration on the common 2 us grid. The Cantera range uses the entire 10 ms reference and never NN values.
N/A: constant or nearly constant Cantera reference (N2 is constant here). —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.